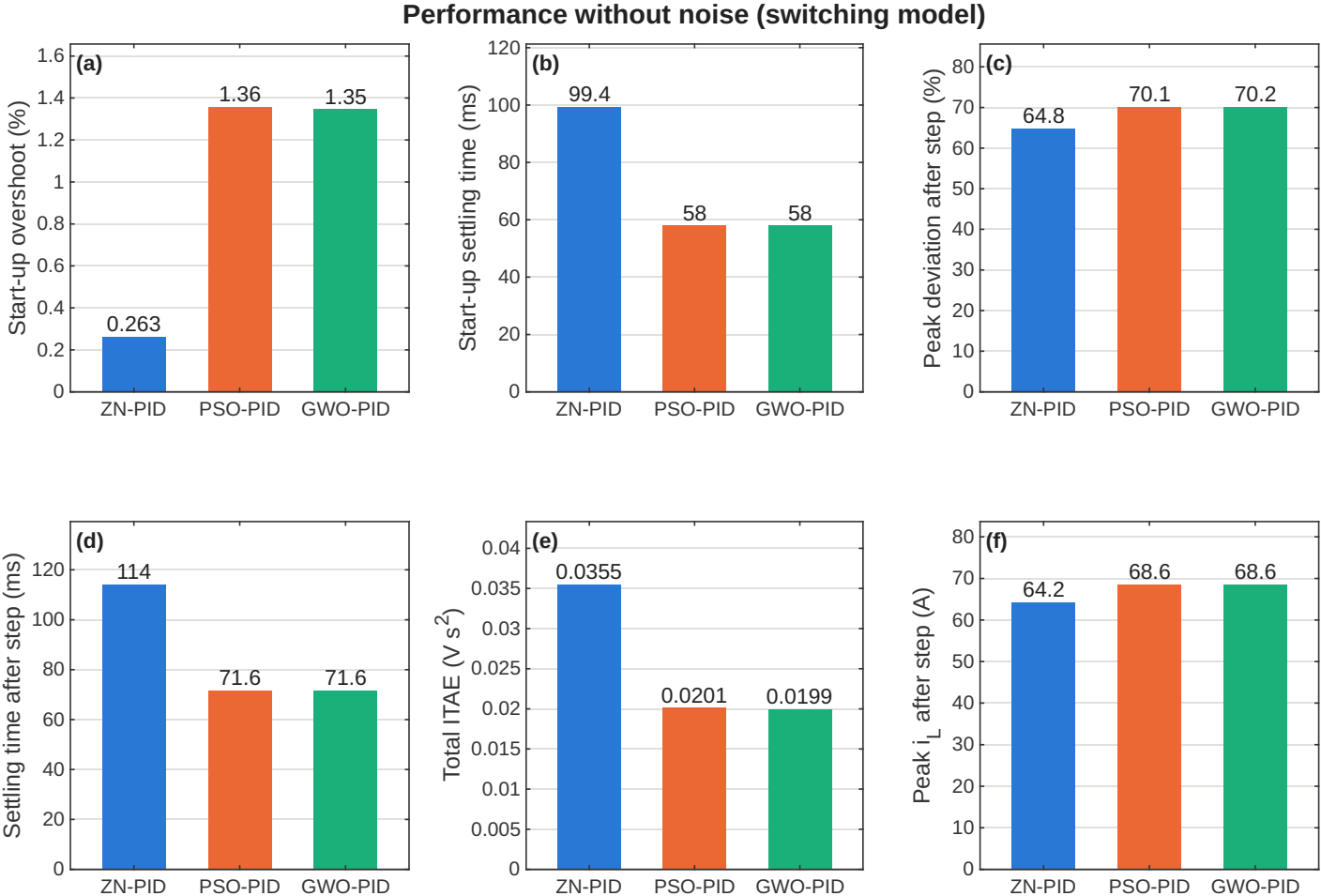
Performance without noise (switching model)
(a)
Start-up overshoot (%)
0.263
1.36
1.35
(b)
Start-up settling time (ms)
99.4
58
58
(c)
Peak deviation after step (%)
64.8
70.1
70.2
(d)
Settling time after step (ms)
114
71.6
71.6
(e)
Total ITAE (V s2)
0.0355
0.0201
0.0199
(f)
Peak iL after step (A)
64.2
68.6
68.6
ZN-PID
PSO-PID
GWO-PID